

## SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS  
COURSE CODE: CSA07

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### COURSE OUTCOME COVERED

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

### Debugging & Optimisation Task

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#### Code of Conduct:

I M. Madhulika (192472188) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Madhulika

## Annexure A

### Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

#### Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

#### Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329:64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

| Destination Network | Next Hop |
|---------------------|----------|
| 192.168.1.0/24      | 10.0.0.2 |
| 192.168.2.0/24      | 10.0.0.3 |
| Default Route       | 10.0.0.1 |

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:



Suggesting IPv4 Network configuration.

IP Address: 192.168.10.65

• Subnet Mask: 255.255.255.192 (126)

• Default Gateway: 192.168.10.1

1) Identify the subnet.

A /26 subnet mask divides the network into blocks of 64 addresses

| subnet            | Address Range |
|-------------------|---------------|
| 192.168.10.0/26   | 0-63          |
| 192.168.10.64/26  | 64-127        |
| 192.168.10.128/26 | 128-191       |
| 192.168.10.192/26 | 192-255       |

The IP 192.168.10.65 belongs to  
subnet: 192.168.10.64/26

2) Network Address, Broadcast Address and Host Range

• Network Address: 192.168.10.64

• Broadcast Address: 192.168.10.127

• Valid Host Range: 192.168.10.65 - 192.168.10.126

3) verify IP Address and gateway.

IP Address

• 192.168.10.65 is a valid host address.

Default Gateway -

- configured Gateway : 192.168.10.1
- Gateway belongs to subnet : 192.168.10.0/26
- workstation belongs to subnet : 192.168.10.64/26

4) configuration errors

- IP address is in 192.168.10.64/26
- Gateway is in 192.168.10.0/26
- Host and gateway are in different subnets

5) Total usable Host Address.

Formula

$$2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = 62.$$

usable Hosts = 62.

Final Answer.

| Item                  | value.                         |
|-----------------------|--------------------------------|
| Subnet                | 192.168.10.64/26               |
| Network Address       | 192.168.10.64                  |
| Broadcast Address     | 192.168.10.127                 |
| Host Range            | 192.168.10.65 - 192.168.10.126 |
| configured IP valid : | Yes                            |
| Gateway valid ?       | No                             |
| correct gateway       | 192.168.10.126                 |
| usable Hosts          | 62                             |



debugging IPv6 configuration  
given

- Device A : 2001:db8::ff00:42:
- Device B : 2001:db8:0:0:1::1/64

2) Determine Network Prefix

Device A 2001:0db8:0000:0000:0000:ff00:0042:8329

Device B 2001:0db8:0000:0000:0001:0000:0000:0001

3)

For /64, the first four blocks represents the network

Device A 2001:db8:0000:0000::/64

Device B 2001:db8:0000:0000::/64

Addressing or Prefix Mismatch

No prefix mismatch exists

Possible communication failure

- Incorrect gateway
- Firewall
- Disabled IPv6
- Neighbor Discovery issues
- Router configuration

Available host Address  
IPv6 host bits

$$128 - 64 = 64$$

Hosts  $2^{64} = 18,446,744,073,709,551,616$

Final Answer

| Item               | value.                                     |
|--------------------|--------------------------------------------|
| Expanded Address A | 2001:0db8:0000:0000:0000:3f00<br>0042:9329 |
| Expanded Address B | 2001:0db8:0000:0000:0001:0000:0000<br>0001 |
| Network prefix     | 2001:db8:0:0::/64                          |
| Same subnet?       | Yes                                        |
| Mismatch?          | No                                         |
| Optimized Address  | 2001:db8:0:0::10 and 2001:db8:0:0::20      |
| Available Host     | $2^{64} = 18,446,744,073,709,551,616$      |

### 3) Analyze and optimize IPv4 subnet Allocation

An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27 and /28 some departments report insufficient IP address, while others have a large number of unused address



## Network

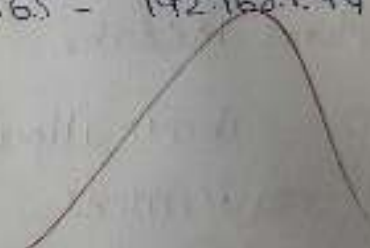
192.168.1.0/24

### Subnet Details.

#### Subnet 1 (126)

- Network Address: 192.168.1.0
- Broadcast Address: 192.168.1.63
- Host Range: 192.168.1.1 - 192.168.1.62
- Total Addresses: 64
- Usable Hosts: 62

#### Subnet 2 (127)

- Network Address: 192.168.1.64
  - Broadcast Address: 192.168.1.95
  - Host Range: 192.168.1.65 - 192.168.1.94
  - Total Addresses: 32
  - Usable Hosts: 30
- 

#### Subnet 3 (128)

- Network Address: 192.168.1.96
- Broadcast Address: 192.168.1.111
- Host Range: 192.168.1.97 - 192.168.1.110
- Total Addresses: 16
- Usable Hosts: 14



## usable and unused IP Addresses

| subnet | usable Hosts | unused (if not fully used) |
|--------|--------------|----------------------------|
| /26    | 62           | Depends on department size |
| /27    | 30           | Depends on department size |
| /28    | 14           | Depends on department size |

## Inefficient Allocation

- small department may waste addresses with a /26 subnet
- Large departments may not have enough addresses with a /28 subnet

## Addressing plan Mistake.

- Fixed size subnet allocation does not match department requirement.
- some departments have insufficient IP addresses while others waste many

## Final Answer

| subnet | Network      | Host Range | Broadcast | usable Hosts |
|--------|--------------|------------|-----------|--------------|
| /26    | 192.168.1.0  | 1-62       | 63        | 62           |
| /27    | 192.168.1.64 | 65-94      | 95        | 30           |
| /28    | 192.168.1.96 | 97-110     | 111       | 14           |

## debugging IPv6 configuration

Given

- Device A : 2001:db8::f00:42:8329/64
- Device B : 2001:db8:0:0:1::1/64

Expanded Address

Device A : 2001:0db8:0000:0000:0000:f00:0042:8329

Device B : 2001:0db8:0000:0000:0001:0000:0000:0001

2) Network Prefix

Both use : 2001:db8::/64

Configuration Error.

There is no prefix mismatch. The problem may be due to

- Incorrect gateway.
- Firewall
- IPv6 disable
- Neighbor Discovery issue.

correct configuration

- Device A : 2001:db8:0:0::1/64
- Device B : 2001:db8:0:0::2/64
- Gateway : 2001:db8:0:0::1



Available Host addresses.

$$2^{128-64} = 2^{64}$$

= 18,446,744,073,709,551,616 addresses

conclusion:

Both devices are already in the same IPv6 subnet  
The communication problem is likely due to  
another network configuration issue

### 5) Optimized Routing Table.

| Destination    | Next Hop |
|----------------|----------|
| 192.168.1.0/24 | 10.0.0.2 |
| 192.168.2.0/24 | 10.0.0.3 |
| Default Route  | 10.0.0.1 |

Ensure all next hop routers are reachable  
and routing information is correct.

Route used if No Match Exists

If no destination network matches the router  
uses the

Default Route  $\rightarrow$  10.0.0.1

inal Answer.

| Item               | value                                                                            |
|--------------------|----------------------------------------------------------------------------------|
| Matching Route     | 192.168.2.0/24                                                                   |
| Next HOP           | 10.0.0.3                                                                         |
| Destination valid? | Yes                                                                              |
| Possible Error     | Incorrect or unreachable<br>next hop, missing return route<br>or interface issue |
| optimized Routing  | verify and maintain correct<br>next hop router and interface<br>status           |
| No matching Route  | Default Route $\rightarrow$ 10.0.0.1                                             |

But